Requirements Engineering (IV)

Training for the IREB CPRE
Foundation Level test

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2h

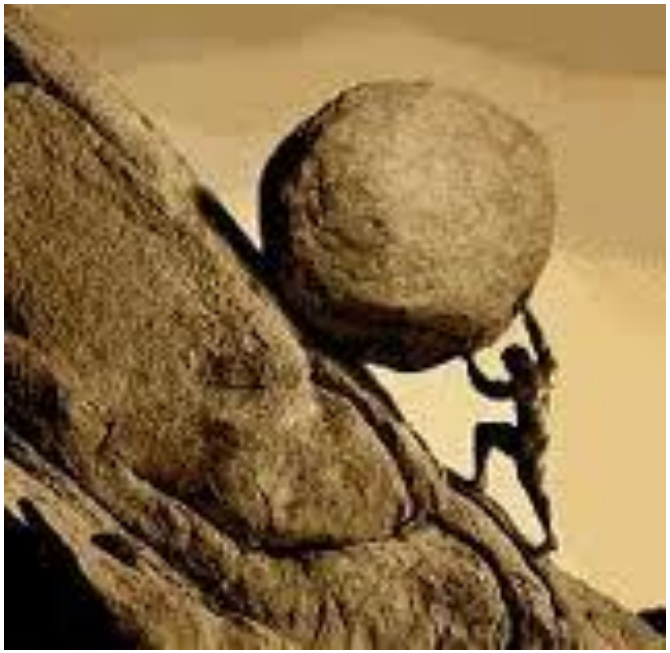
EU 4 – REQUIREMENTS DOCUMENTATION

Terms

- Requirements Document,
- Requirements Specification

A few myths that hurt

- “requirements are known”
- “requirements don’t change”
- “only technical requirements are to be described”
- “only features are to be described”
- “requirements must focus of the solution”
- “requirements are only about the software”
- “the requirements are specified by the use cases”
- “the requirements descriptions are independent from other descriptions”



Requirements exchange models

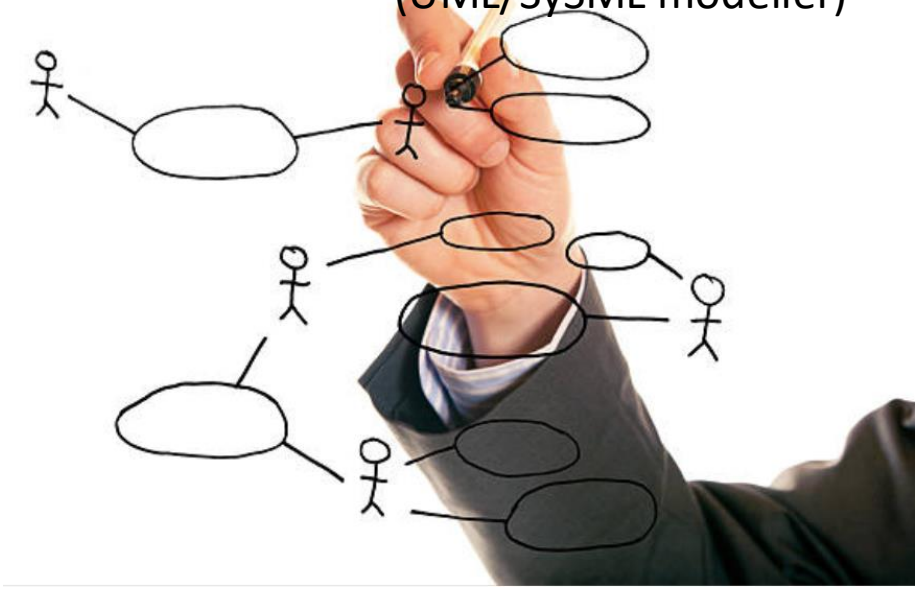


Oral communication



Written requirements
(Excel, Word)

Formal requirements models
(UML/SySML modeller)



Different ways to document requirements

Oral

Most frequent, typically the first form of communication

Difficult to manage (traceability, configuration managements, etc)

Easy to make mistakes

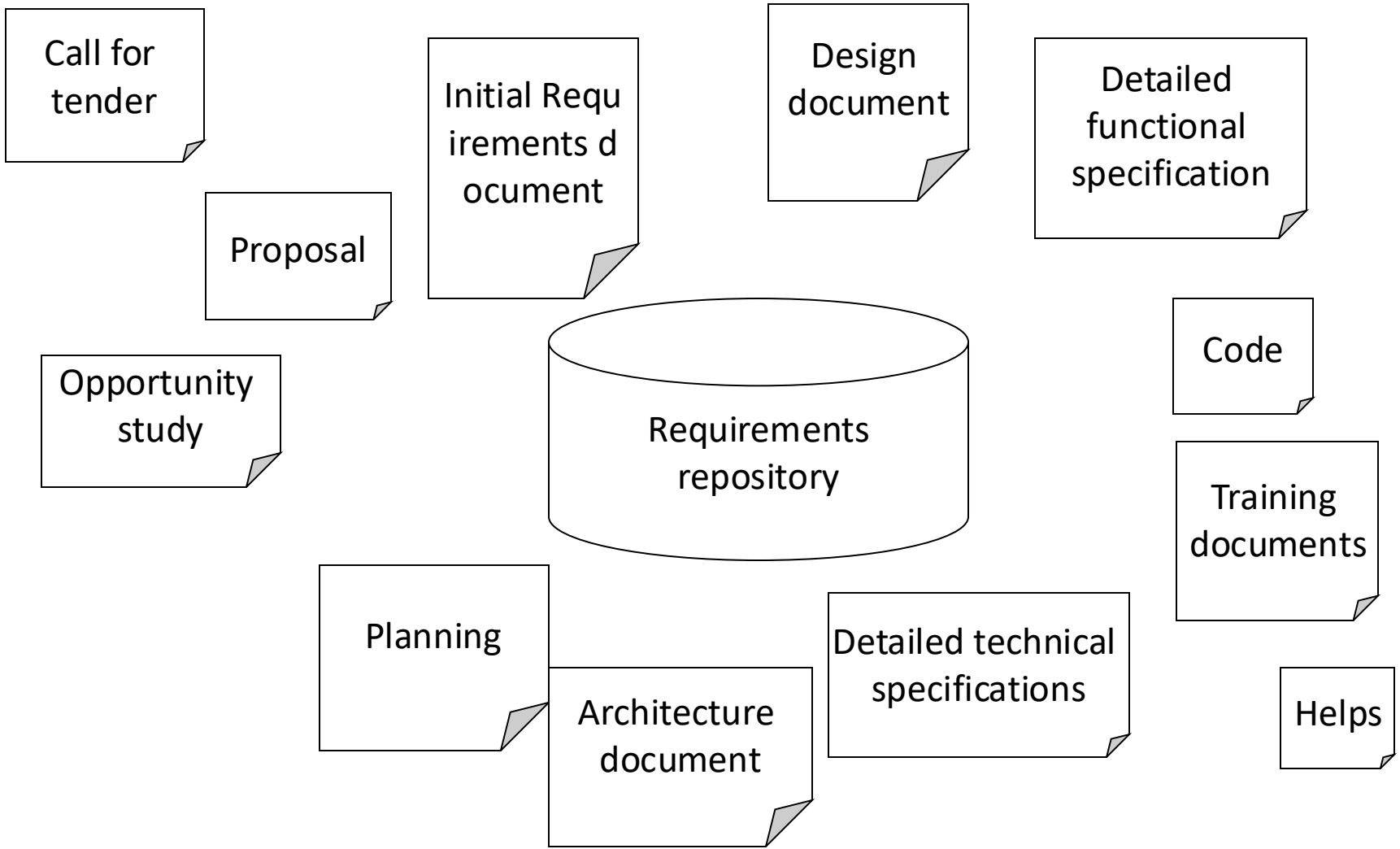
As a written data

- Typically Excel tables and requirements templates
- If « known », every partner has its own definition and has to interprate the other ones
- If « negociated », the meaning of properties is shared and communication is easier

As an information

- Typically repository and formal requirements models
- Ability to communicate requirements « in their context »
- Facilitates synchronization
- Improves traceability

Requirements documentation « as an information system »



Requirements documentation: two dimension

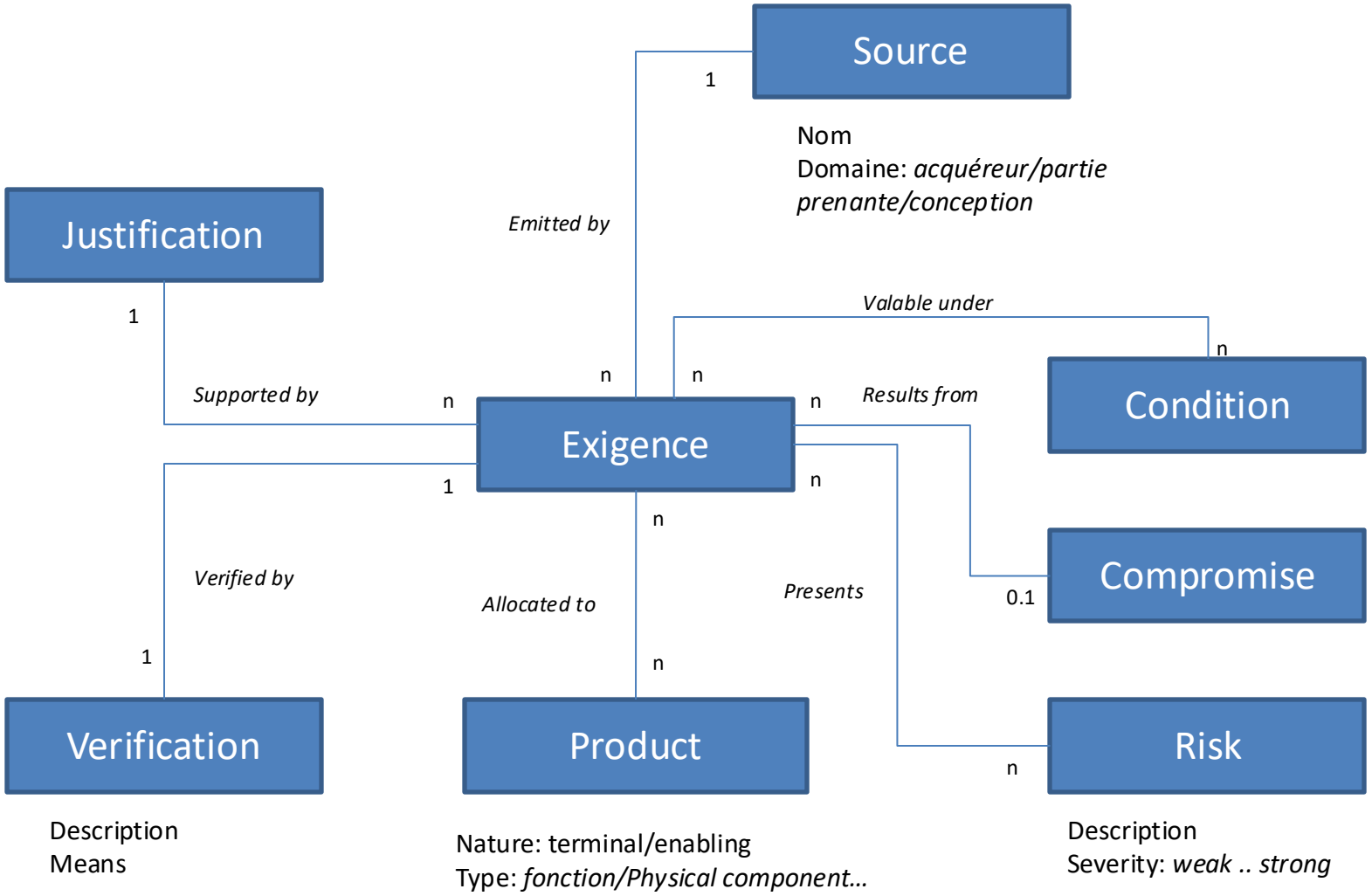
External structure

- The different kinds of documents or containers with their mission
- Relationships between them

Internal structure

- The different kinds of information (goals, context, problem statements, solution statements, purpose and motivations, etc) contained in the documents
- The internal organisation of the document

Requirements repository structure (Source : AFIS)



Good practice (1)

Use standards for structuring reqts documents

- Easy way to structure documentation
- Provides a reference structure for requirements-related artefacts (goals, scenarios, FRs, quality attributes, constraints)
- Benefits of the reference struture:
  - Better requirements authoring
  - Finding the blanks and filling the gaps
  - Training stakeholders
  - Provides easy acces to information (eg for tests)
  - Reuse

Intrenal structuring ofan SRS : IEEE 830

1 Introduction

- Purpose of the requirements document
- Scope of the product
- Definitions, acronyms and abbreviations
- References
- Overview of the remainder of the document

2 General description

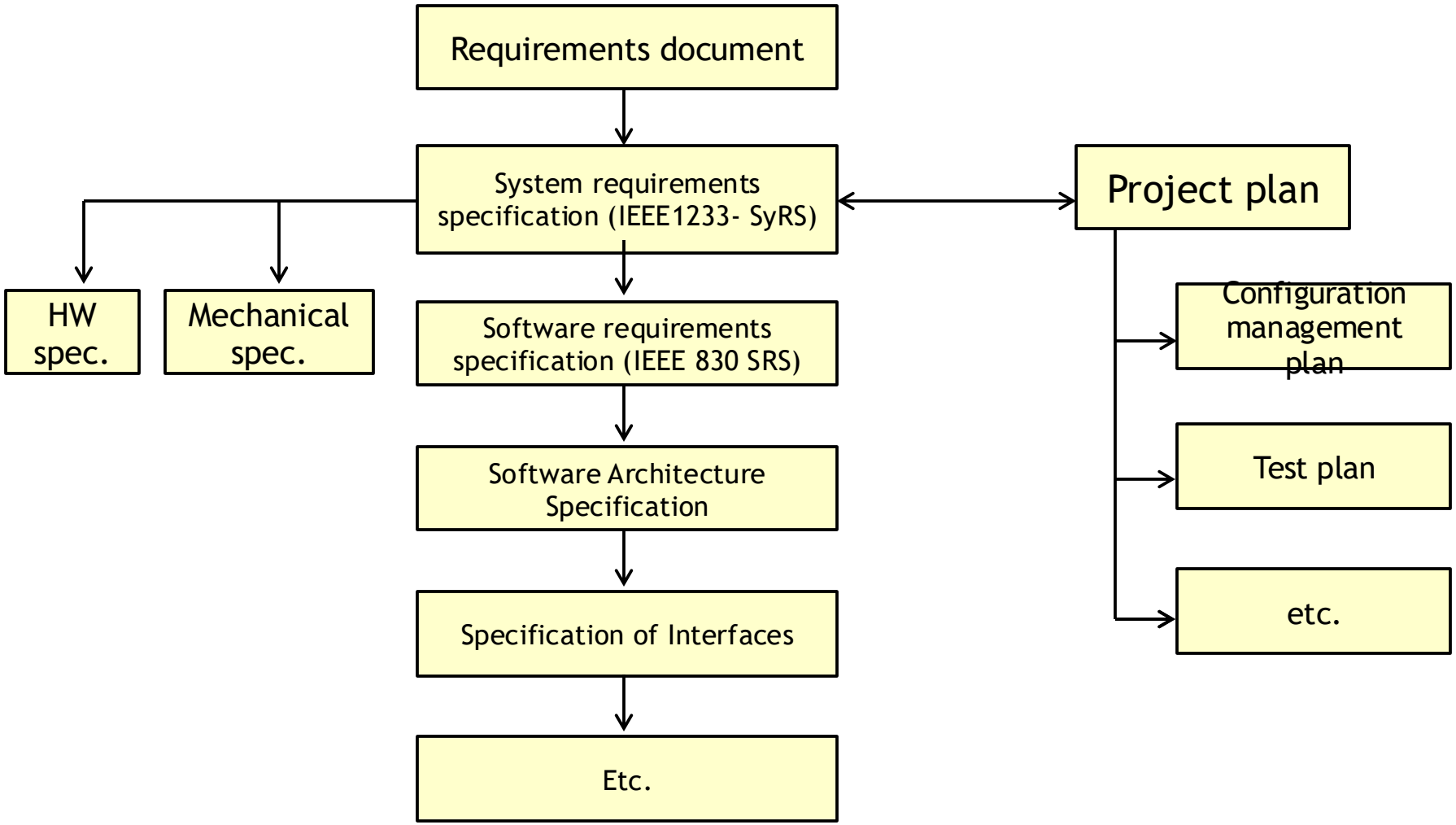
- Product perspective
- Product functions
- User characteristics
- General constraints
- Assumptions and dependencies

3 Specific requirements

- functional requirements
- non-functional requirements
- interface requirements

...

External structuring of SRS : IEEE 1233 (SyRS)



Quality criteria for a requirements document



Complete



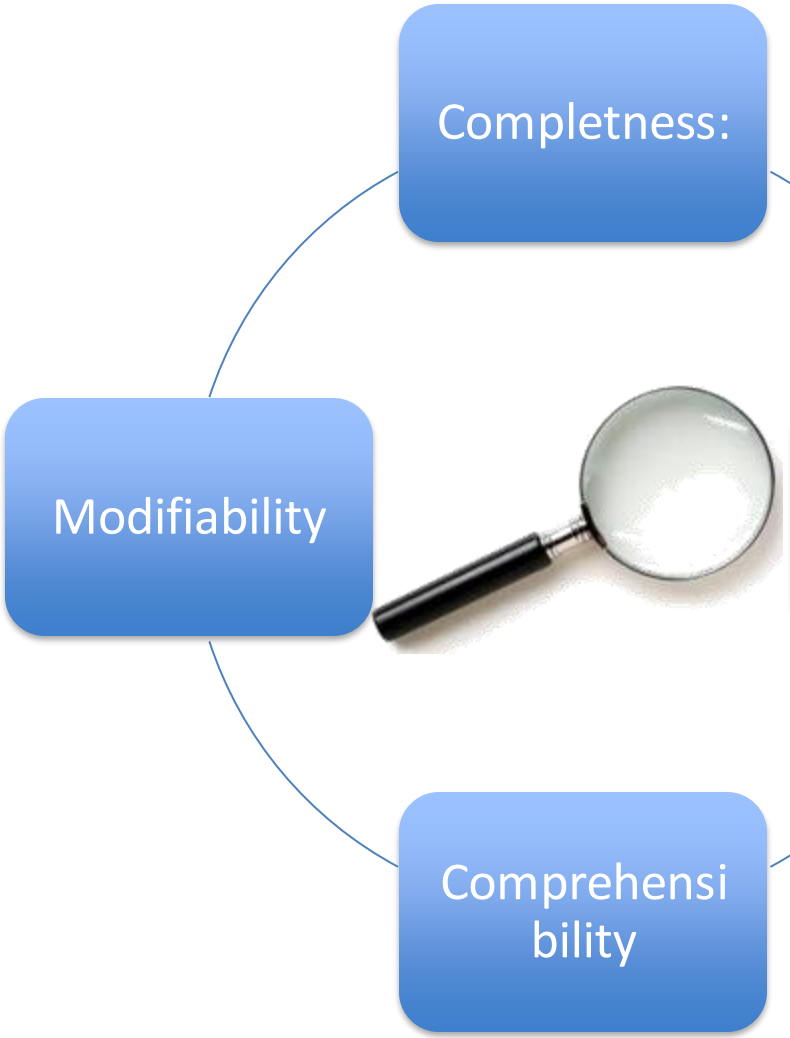
Consistent



Clear, modifiable and extensible structure

Good practice (2)

Organize a **review**



Other verifications

- **Unique**: no redundancy
- **Annotated**: each requirements has all the needed information (context, priority, release, etc)
- **Organized**: a place for each requirements and each requirements at its place
- **Design independentant** : solutions described elsewhere
- **Traced** : author, source and justification are documented

Good practice (3)

Define a **dictionnary/glossary**

Terms

Acronyms



Definitions

Shared understainding of business terminology

Synonyms

Good practice (4)

Good practice

- provide a simple, precise and intelligible definition for each relevant term in the field, and a list of accepted synonyms.
- choose a term for an object, and stick to it (e.g. "handset", "mobile", "mobile phone", "Terminal Equipment", "TE", "telephone"..)
- The glossary should be "appropriated" by all stakeholders.
- and accessible to all (e.g. in the form of a "wiki").

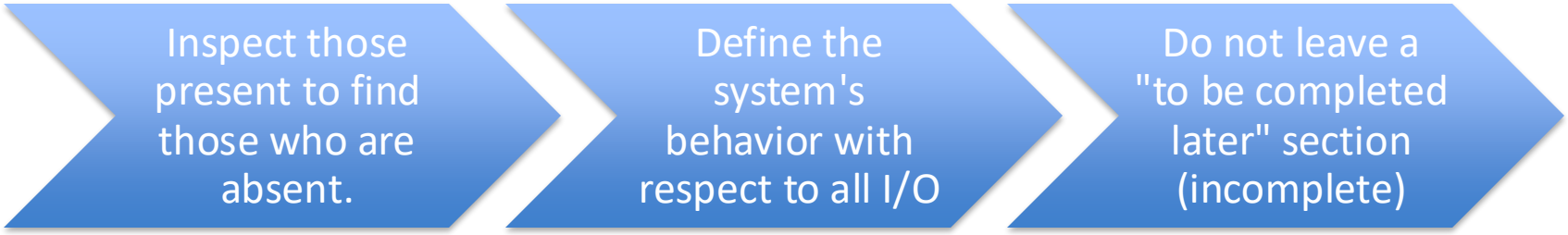
[Lamsweerde 2009] §3.1.1

Inconsistencies and **conflicts** risks

- Terminology clash due to **synonymes**
  - Eg: « base screen » =« idle screen »
  - Eg: « main screen » = « home screen »
- Designation clash due to **homonyms**
  - Eg : « ring tone » = « sms bip » + « wake up tone » + « telephone ring », etc

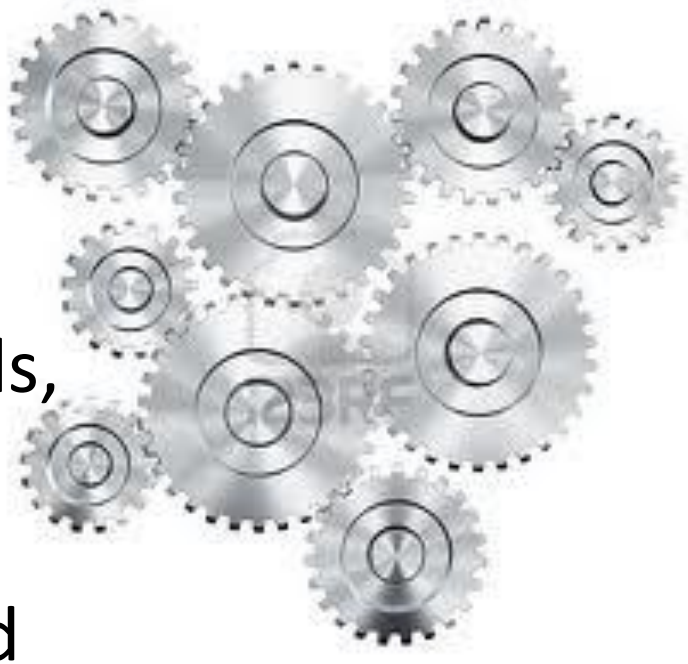
Complete

- The specification contains everything the product is supposed to do
- The requirements defined represent exactly the needs of the users
- It is difficult to verify completeness:



Complete with elicitation techniques

- Interviews
- Questionnaires
- Focus group
- Brainsoriming
- Document review (standards, manuals, laws, etc.)
- Prototyping
- Field observation (quick and dirty ethnography)
- etc



Good practice (5)

Complete by exploring in both directions

Bottom up

- Questioning the existing situation and the recommended solutions
- Identify problems
- Abstracting from the details
- Generalize at the domain level
- Ask the question of why (of the project, the system, the organization, etc.)

Top down

- Using analogical reasoning
- Being creative involves "inventing" requirements as needed
- Identify the parameters of the solutions already identified and decline them
- Recombine

Good practice (6)

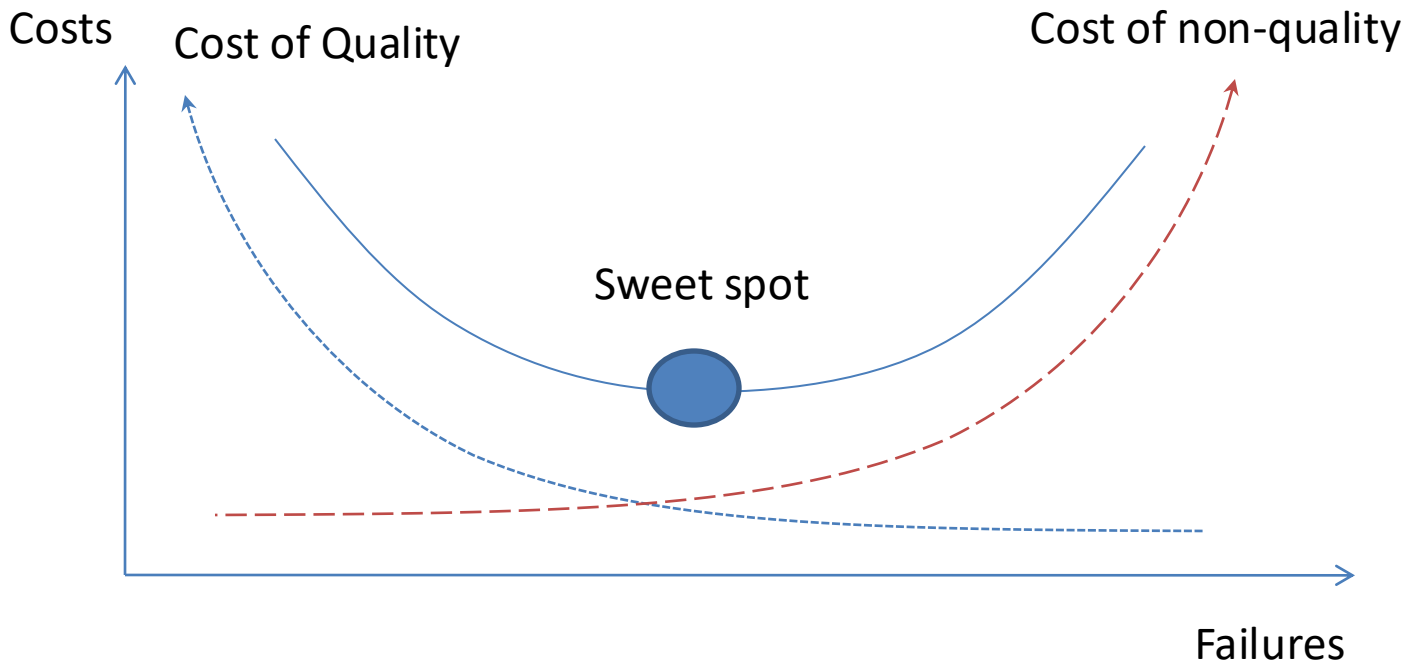
Keep quality criteria in mind

- Standard IEEE 830
  - Unambiguous
  - Complete
  - Consistent
  - Verifiable
  - Editable
  - Traceable
  - Usable

- Other
  - Annotated
  - Real
  - Comprehensible
  - Expressive
  - Testable
  - Building



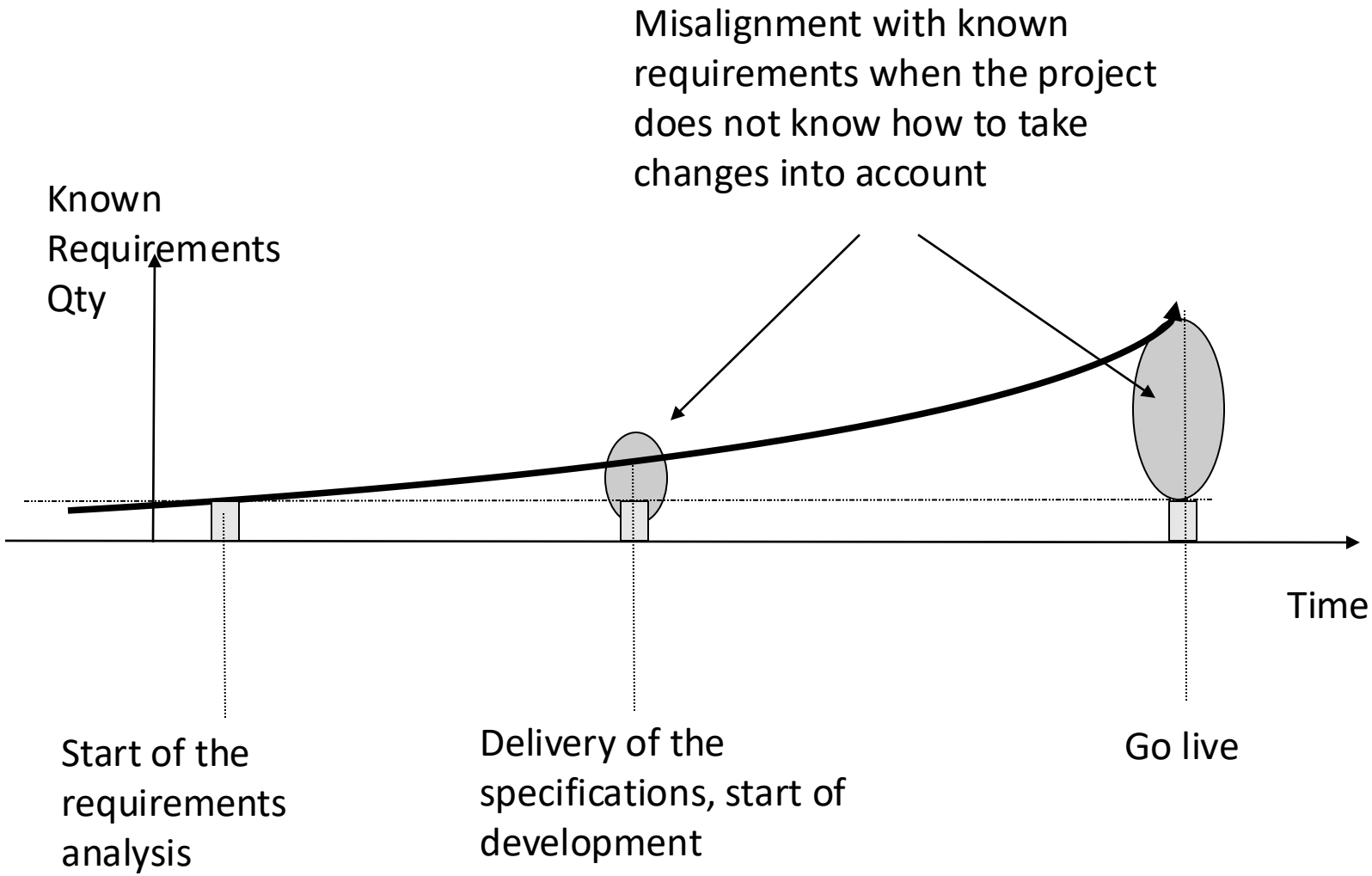
No need to overdo it!



Coût de la fiabilité des systèmes

Source: « Le marketing des SI », Guillaume Puiseux, Eyrolles 2008

Stability issue: the « requirements drift »



Takaway: Design your requirements documentation!

- Many standards exist (e.g. IEEE 830-1998 SRS and IEEE 1233-1998 SysRS)
- Simple way to structure documentation
- Goals, scenarios, and functional requirements, quality requirements, constraints, etc.
- The benefits of these reference structures:
- Writing (clarifies expected information)
- Search for gaps (completeness)
- Staff training
- Access to information (e.g. testing) and reuse

